UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 010 · Problems on maxima and minima for a function of two variables · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S010·8	Deck 010, slide 8 (rendered .pptx)
T010a 00:07:38	AV-summary transcript “Extremas of functions continued” (older lecture), segment start
T010b 00:03:12	AV-summary transcript “Problems on Maxima and Minima for a function of two variables” (newer lecture)
CW U1 P29	Unit 1 class-work problem 29
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c
ESA Jan-24 s2 1a	Past paper: Jan 2024 set 2, question 1.a

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2. 1 · The method, and the ESA cubic functions
3. 3 · Applied problems: eliminate one variable first
4. Part B · Symbol glossary

A Formula sheet

1 The method, and the ESA cubic functions

E1.1 · Stationary points ESA ★★★

$$f_x = 0, \qquad f_y = 0 \quad \text{(solve simultaneously)}$$

GIVES	All candidate points.
APPLIES	Every problem, first step.
NOTE	Factorise: $6xy - 30y = 6y(x-5) = 0$ gives the cases $y = 0$ or $x = 5$.

S010·7 · T010b 00:01:23 · T010b 00:01:40

E1.2 · The test

ESA ★★★

$$r = f_{xx}, \quad s = f_{xy}, \quad t = f_{yy}$$
$$rt - s^2 > 0 :\ r < 0 \text{ max},\ r > 0 \text{ min}; \qquad rt - s^2 < 0 :\ \text{saddle}; \qquad rt - s^2 = 0 :\ \text{doubtful}$$

GIVES The nature of each stationary point.

APPLIES At each point separately.

NOTE Restated from class 009.

S010·8 · T010b 00:03:12 · T010b 00:04:02

E1.3 · Hessian of the ESA cubic

ESA ★★★

$$f = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x : \qquad rt - s^2 = 36\left[(x-5)^2 - y^2\right]$$

GIVES $rt - s^2$ at every point at once.

APPLIES Dec-23 1c, Jan-24 s1 1a.

NOTE $r = t = 6x - 30$, $s = 6y$. Slide 010·8 drops the 36 on y^2.

S010·8 · T010b 00:03:30

3 Applied problems: eliminate one variable first

E3.1 · Eliminating with a sphere

ESA ★★

$$x^2 + y^2 + z^2 = a^2 \ \Rightarrow\ z^2 = a^2 - x^2 - y^2, \qquad T = kxyz^2 = kxy(a^2 - x^2 - y^2)$$

GIVES A function of x, y only.

APPLIES Quantity contains z^2 and the point lies on a sphere.

NOTE Unit sphere: $a = 1$.

S010·3 · S010·9 · T010a 00:01:00

E3.2 · Open box of volume V

ESA ★★

$$A = xy + 2yz + 2zx, \quad xyz = V \ \Rightarrow\ A = xy + \frac{2V}{x} + \frac{2V}{y}$$

GIVES Surface area (metal used) of a box with no top, base $x \times y$.

APPLIES Least material for a given volume.

NOTE Minimum at $x = y = \sqrt[3]{2V}$, $z = \frac{x}{2}$ (CW P34). The slide's $A = 2xy + 2yz + zx$ is not the area of an open box.

ADDED FOR CLARITY S010·5 · CW U1 P34

B Symbol glossary

Symbol	Name	Meaning	Appears in
$p,\ q$	First partials	$p = f_x$, $q = f_y$ (lecture T010b)	E1.1
$r,\ s,\ t$	Second partials	f_{xx}, f_{xy}, f_{yy} at a point	E1.2, E1.3
$rt - s^2$	Hessian	Decides extremum or saddle	E1.2, E1.3
T	Temperature	$400xyz^2$ or $kxyz^2$; not the second partial t	E3.1
k	Constant	Positive constant in $T = kxyz^2$	E3.1
a	Radius	Sphere $x^2 + y^2 + z^2 = a^2$	E3.1
A	Surface area	Metal used by the open box	E3.2
V	Volume	Fixed capacity $xyz = V$ (10 m³)	E3.2

Remark. The slides write $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$ and test the sign of $rt - s^2$; other books call it the discriminant D or the Hessian. The lecture of T010b also names the first partials $p = f_x$ and $q = f_y$. Do not confuse $t = f_{yy}$ with the temperature T of slides 010·3 and 010·9.